

Project Plan: Predictive Analytics for Customer Churn

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Overview

Introduction and Dataset

In the current landscape of 2026, the telecommunications and SaaS industries have reached a point where customer acquisition is significantly more expensive than customer retention. Churn has become one of the most important metrics that determines whether a company really performing or just surviving. Traditionally, businesses have operated in a reactive mode, only realizing a customer is unhappy once they have already initiated a cancellation. I am interested in learning how I can use advanced predictive modeling techniques to identify this shift in customer behavior before it happens. By mastering these tools, I want to understand how to move from simply observing a loss to proactively stopping it through data-driven intervention.

The real value in this project is not just in creating an algorithm that spits out a probability score. My goal is to build a model that actually identifies the key "pain points" that drive a customer to leave, such as specific contract types or recurring service glitches. By uncovering these underlying triggers, a business can transition from a broad, generic retention strategy to one that is highly effective and personalized. If we can understand why a customer is frustrated, we can fix the root cause rather than just offering a temporary discount to stay.

The Problem Statement

The goal of this project is to build a predictive model that identifies which customers are at high risk of leaving a telecom provider. Rather than just providing a simple probability score, I want to uncover the specific drivers of churn, such as contract types or service glitches. By identifying these early red flags, the business can intervene with targeted retention efforts that actually address the reasons why a customer is unhappy. I will be following the CRISP-DM framework—Cross-Industry Standard Process for Data Mining—to ensure that the technical work stays aligned with these business goals.

The Initial Dataset

The primary data for this project is the Telco Customer Churn dataset sourced from Hugging Face. This dataset contains information on 7,043 customers, providing a solid foundation for statistical analysis. The features are divided into three main categories: demographic details (gender, senior citizen status, partners), account information (tenure, contract type, paperless billing), and service usage (internet type, online security, tech support). I plan to spend the first two weeks specifically on Phase 1, "Getting into the Data," to find early correlations, such as whether month-to-month contracts are significantly more prone to churn than two-year commitments. While this is a comprehensive starting point, I will also be pulling in external economic indices or regional service outage data if the initial features do not show a strong enough signal to predict churn accurately.

Modeling and Evaluation

Choice of Models

For this project, I plan to move beyond a single algorithm and instead conduct a robust "head-to-head" competition between several distinct modeling architectures. I will start with a baseline Logistic Regression to establish a performance "floor" and understand the linear relationships between features like monthly charges and churn. However, to capture the more complex "pain points" in the customer journey, I will implement and compare the following classifiers:

Random Forest and XGBoost:

These ensemble methods are my starting point for tree-based learning, as they are excellent at identifying non-linear interactions between variables, such as how a specific contract type might only trigger churn when paired with a lack of tech support.

Support Vector Machines (SVM):

I want to test if a high-dimensional hyperplane can better separate churners from loyal customers, particularly when dealing with the many service-related features in the Telco dataset.

K-Nearest Neighbors (KNN):

This will allow me to explore "customer similarity." I want to see if customers who churn tend to cluster together in the feature space, which could provide intuitive insights for the business.

Neural Networks (MLP):

Finally, I will test a Multilayer Perceptron to see if a deep learning approach can uncover hidden patterns that traditional statistical models might miss.

Ultimately, I will not rely on a single model. I plan to evaluate the performance of each and choose the best-performing models based on their specific classifier metrics and their ability to generalize to unseen data.

Evaluation Strategy:

Once these models are trained, I will evaluate them using a Confusion Matrix and AUC-ROC curves.

However, as I noted in my initial plan, missing a customer who is about to leave is much more expensive than a "false alarm". Therefore, I will prioritize Recall as my primary metric to ensure the model captures as many at-risk customers as possible. I will also use F1-Scores to balance precision and recall, ensuring the model remains practical for business use. To prevent the models from overfitting the data, I will implement K-Fold Cross-Validation, which ensures that my selection of the "best" model is based on stable, repeatable performance across different subsets of the data.

Goals, Risks, and Contingencies

What I Hope to Learn

Ultimately, I hope to identify the features that are really important, which drive a customer to walk away. I want to learn if it is a lack of tech support or if the month on month contract structure is the primary

driver of dissatisfaction. By uncovering these specific pain points, I can provide a roadmap for where a business should invest its resources to improve the customer experience. I also want to calculate a rough ROI. If we can catch even 15% of churners early, what does that actually save the business in the long run? This predictive analysis and the later cost benefit analysis will provide the real-world impact regarding how impactful and actionable a business strategy is.

Risks and Ethics

One major risk is data bias. If the training data is heavily weighted toward one specific region or demographic, the model might unfairly flag certain groups as "high risk," leading to biased business decisions. There is also the ethical concern of Data Privacy. I must ensure that any personally identifiable information is handled securely and that the model's predictions are used for positive retention rather than predatory pricing or excluding certain groups from promotions. Furthermore, the risk of "model drift" exists, where a model that works today might become inaccurate as market conditions change in the future.

The Contingency Plan

Data science is rarely a straight line. If the initial data exploration in Phase 1 reveals that the features are too "noisy" or the signal is too weak to make reliable predictions, my contingency plan is to pivot toward heavy feature engineering. This would involve creating new metrics, like "monthly charges relative to tenure," or "ratio of tech support calls to total months," to see if those reveal deeper patterns. If that still does not work, I will supplement the project with a different, more robust open-source telecom dataset to ensure the research goals are still met. I have coordinated with my peer reviewer, Amanda Wyse, to swap

documentation early in the process to ensure my logic is sound before I commit too deeply to the modeling phase.

Closing

This proposal serves as the foundation for the final project submission. I am following the CRISP-DM framework to keep the workflow organized as the data gets messy. While my current hypothesis focuses on service quality and contract types, I acknowledge that the findings during the data prep and modeling phases might take this project in a completely different direction. This is a "to be continued" process where the data will ultimately dictate the final strategy.

Phase 2

Will I be able to answer the questions I want to answer with the data I have?

Yeah, I'm confident about this. My dataset gives me everything I need to figure out what drives churn, predict it accurately, and compare models. I've got contract type, tenure, monthly charges, internet service type, payment method, and offers all in there. My feature importance analysis from the Random Forest already shows which ones matter most. The only thing I don't have is behavioral trends over time, like whether a customer's usage is declining month after month. That would be nice to have for predicting churn earlier, but it's not essential. My current data feels solid for what I'm trying to do.

What visualizations are especially useful for explaining my data?

I think I've actually chosen my visualizations really well. The churn rate charts by contract, internet type, offer, and payment method are super clear for business people because I can immediately see which segments have the highest risk. The histogram showing tenure and monthly charge distributions split by whether someone churned or not is helpful because it reveals the pattern that early tenure, high bill

customers are my problem. My feature importance bar chart directly shows what my model learned. The ROC curves let me see which models discriminate best. The clustering analysis with churn rate by segment ties everything together. I will add more from the list I have provided in part 1. Also, I will show a confusion matrix as a heatmap for visual impact, or a plot showing how precision and recall change with different prediction thresholds.

Do I need to adjust my data and/or driving questions?

As of now the data feels sufficient and my questions do well framed but I am still learning and you could provide or add comments if I need to add more context to them. I'm trying to ask: which attributes drive churn, how accurate can I predict, which model is best. These questions directly support business decisions about retention. I think the framed questions are good and need no change per my understanding. On the data side, one limitation I should acknowledge is that I have a single snapshot from one telecom provider.

Do I need to adjust my model/evaluation choices?

The approach feels sound. Using class weight balancing is the right call for my 73.5 vs 26.5 split. Prioritizing ROC-AUC, F1, and Recall instead of just accuracy shows I understand imbalanced classification. Testing seven different models from dummy baseline through ensemble methods shows good rigor. My GridSearchCV for tuning Random Forest is appropriate. If I want to take it one step further before deployment I could test different decision thresholds and see how precision and recall trade off at each one, then pick the threshold that matches my business cost of false negatives versus false positives. But that's optional and more of a production readiness thing.

Are my original expectations still reasonable?

Absolutely. I set out to identify the specific drivers of churn instead of just getting a probability score, and I am working on that. I will add more information in coming weeks to provide better data driven decision.

Contract type, tenure, monthly charges, and internet service type all emerged as key drivers. I wanted a model that beats a simple baseline, and my ensemble methods are doing better than the dummy classifier on ROC-AUC and F1. I wanted to segment customers into actionable groups, and my K-Means clustering gives me clear segments with different churn profiles. As of now, everything I aimed for is working. The one realistic caveat is that with 73 percent non-churn and 27 percent churn, I'm never going to achieve super high precision at the same time as super high recall. That's just the nature of imbalanced data. But the chosen metrics acknowledge that tradeoff and account for the business reality that missing a churner is more costly than flagging a false alarm.

Phase 3

Explain your process for prepping the data

I prepared the data by first loading training and test datasets separately so I could avoid data leakage from the start. Then I profiled the target variable and confirmed class imbalance, which helped me decide to prioritize recall, F1, and ROC-AUC instead of accuracy alone. After checking missing values, I removed post-outcome fields like churn reason and churn category because those would leak future information into the model. For the remaining missing values, I used business-meaningful imputations (for example, filling missing offer and internet type with labels that reflect real customer states). Finally, I split features and target, one-hot encoded categorical variables, and aligned train/test columns to ensure both datasets had the same feature structure before modeling.

Build and evaluate at least one model

I started with a Dummy Classifier as a baseline and used it to prove that a naive model can look acceptable on accuracy while still failing to identify churn cases. After that, I trained a Logistic Regression model, followed by a class-balanced Logistic Regression to better handle minority churn cases. I also

trained ensemble models such as Random Forest (with GridSearchCV tuning), Extra Trees, Gradient Boosting, and AdaBoost. I evaluated model performance using confusion matrices, classification reports, ROC curves, and metrics including accuracy, precision, recall, F1, and ROC-AUC. This gave me both a baseline comparison and a stronger set of candidate models for churn prediction.

Interpret your results

The results show that model quality improves substantially beyond the dummy baseline, confirming that the feature set captures meaningful churn signals. Ensemble models, particularly Random Forest and Gradient Boosting, delivered the strongest overall discrimination based on ROC-AUC and F1 patterns. Feature insights and EDA tell a consistent story: churn is more likely among month-to-month customers, shorter-tenure customers, and customers with higher monthly charges or certain service/payment profiles. In other words, churn is not random; it clusters around specific behavioral and contract-related conditions. This makes the model output interpretable and actionable rather than just predictive.

Begin to formulate a conclusion/recommendations

A practical conclusion is that churn can be predicted early enough to support proactive retention, and that targeted intervention should focus on high-risk segments rather than broad campaigns. I would recommend prioritizing retention outreach for month-to-month, early-tenure, and higher-bill customers, while tailoring offers to the risk factors shown in the analysis. From a modeling perspective, I would deploy one of the top ensemble models, then tune the decision threshold based on business costs of false negatives versus false positives. I would also monitor for model drift and retrain on a schedule, since customer behavior and market conditions can change over time.

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